SULIT

SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 2

Name	Chew Chiu Xian
IC No. / Matric No.	A23CS0061
Year / Program	2/SECPH
Section	02
Lecturer Name	Dr. Seah Choon Sen
Case Study	Saving Management in KADA System
System Name	Koperasi KADA Online Platform
Group Name	Tech EcoPioneer

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SECTION C (SQL AND INTERFACE IMPLEMENTATION)

3.1 SQL Statement (DDL & DML)

a) DDL Statements

1. Create Table for Members

```
CREATE TABLE Members (
    memberID INT PRIMARY KEY,
    memberName VARCHAR(255),
    memberIC VARCHAR(22),
    memberGender ENUM('Male', 'Female'),
    memberReligion VARCHAR(50),
    memberRace VARCHAR(50),
    memberMaritial_status ENUM('Single', 'Married', 'Divorced', 'Widowed'),
    memberPFNo INT,
    memberSalary DECIMAL(10,2),
    memberPosition VARCHAR(100),
    memberGrade VARCHAR(50),
    memberStatus ENUM('Pending', 'Lulus', 'Tolak'),
    applicationDate TIMESTAMP,
    approvalDate TIMESTAMP
);
```

Output:

memberID	memberName	memberIC	memberGender	memberReligion	memberRace	memberMaritial_status	memberPFNo	memberSalary	memberPosition	memberGrade	memberStatus	applicationDate	approvalDate

2. Create Table for Accounts

```
CREATE TABLE Accounts (
    accountID INT PRIMARY KEY,
    memberID INT,
    created_at TIMESTAMP,
    FOREIGN KEY (memberID) REFERENCES Members(memberID)
);
```

Output:

accountID	memberID	created_at

b) DML Statements

1. Insert Data into Members

```
INSERT INTO Members (memberID, memberName, memberIC, memberGender, memberReligion, memberRace, memberMaritial_status, memberPFNo, memberSalary, memberPosition, memberGrade, memberStatus, applicationDate, approvalDate)

VALUES (1, 'JOHN DOE', '890101-01-1234', 'Male', 'Christian', 'Malay', 'Single', 1001, 3500.50, 'Clerk', 'A1', 'Lulus', '2025-01-01', '2025-01-10');
```

Output:

memberID	memberName	memberIC	memberGender	memberReligion	memberRace	memberMaritial_status	memberPFNo	memberSalary	memberPosition	memberGrade	memberStatus	applicationDate	approvalDate
1	JOHN DOE	890101-01-1234	Male	Christian	Malay	Single	1001	3500.5	Clerk	A1	Lulus	1/1/2025	10/1/2025

2. Insert Data into Accounts

```
INSERT INTO Accounts (accountID, memberID, created_at)

VALUES (101, 1, '2025-01-15');
```

Output:

accountID	memberID	created_at
101	1	15/1/2025

3. Join Query 1: Retrieve Member Details with Savings Accounts

```
SELECT Members.memberID, Members.memberName,
SavingsAccount.accountNumber, SavingsAccount.balance

FROM Members

JOIN Accounts ON Members.memberID = Accounts.memberID

JOIN SavingsAccount ON Accounts.accountID = SavingsAccount.accountID;
```

Output:

memberID	memberName	accountNumber	balance
1	JOHN DOE	SA001	5000
2	JANET DOE	SA002	800

4. Update Member's Salary

```
UPDATE Members

SET memberSalary = memberSalary + 500.00

WHERE memberID = 1;
```

Output:

memberID	memberName	memberIC	memberGender	memberReligion	memberRace	memberMaritial_status	memberPFNo	memberSalary	memberPosition	memberGrade	memberStatus	applicationDate	approvalDate
1	JOHN DOE	890101-01-1234	Male	Christian	Malay	Single	1001	4000.5	Clerk	A1	Lulus	1/1/2025	10/1/2025

5. Delete a Recurring Payment

```
DELETE FROM RecurringPayments

WHERE paymentID = 201;
```

Output:

paymentID	accountID	PaymentAmount	last_deduction_date	paymentmethod	next_deduction_date

6. Join Query 2: Get Members with Savings Greater than a Specific Amount

```
SELECT Members.memberID, Members.memberName, SavingsAccount.balance

FROM Members

JOIN Accounts ON Members.memberID = Accounts.memberID

JOIN SavingsAccount ON Accounts.accountID = SavingsAccount.accountID

WHERE SavingsAccount.balance > 1000.00;
```

Output:

memberID	memberName	balance
1	JOHN DOE	5000

7. Insert a New Transaction Record

```
INSERT INTO Transactions (TransactionID, AccountID, tAmount, type, PaymentMethod, referenceNo, description, created_at)

VALUES (302, 101, 1000.00, 'deposit', 'fpx', 'REF002', 'One-time savings deposit', '2025-01-16');
```

Output:

TransactionID	AccountID	tAmount	type	PaymentMethod	referenceNo	description	created_at
302	101	1000	deposit	fpx	REF002	One-time savings deposit	16/1/2025

8. Join Query 3: Get Member Details with Savings Accounts based on Member ID

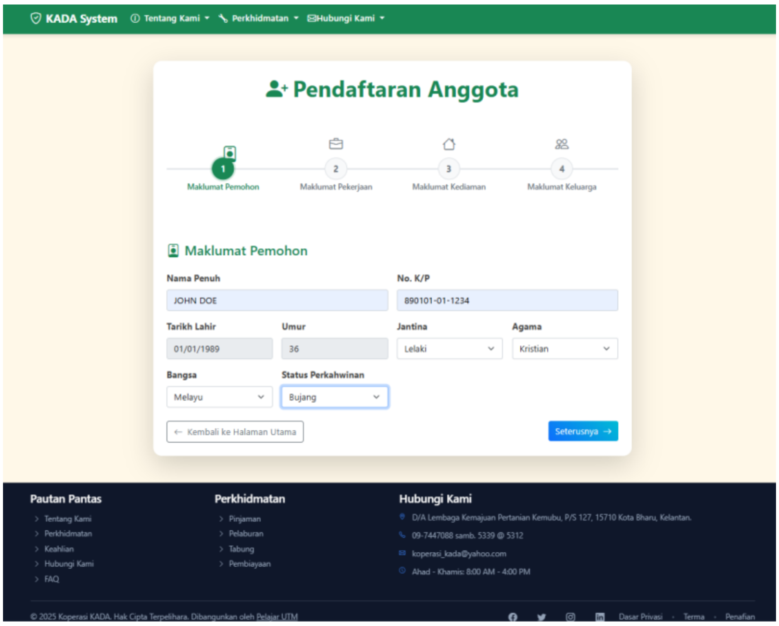
```
SELECT Members.memberID, Members.memberName, SavingsAccount.accountNumber, SavingsAccount.balance

FROM Members

JOIN Accounts ON Members.memberID = Accounts.memberID

JOIN SavingsAccount ON Accounts.accountID = SavingsAccount.accountID

WHERE Members.memberID = 2;
```

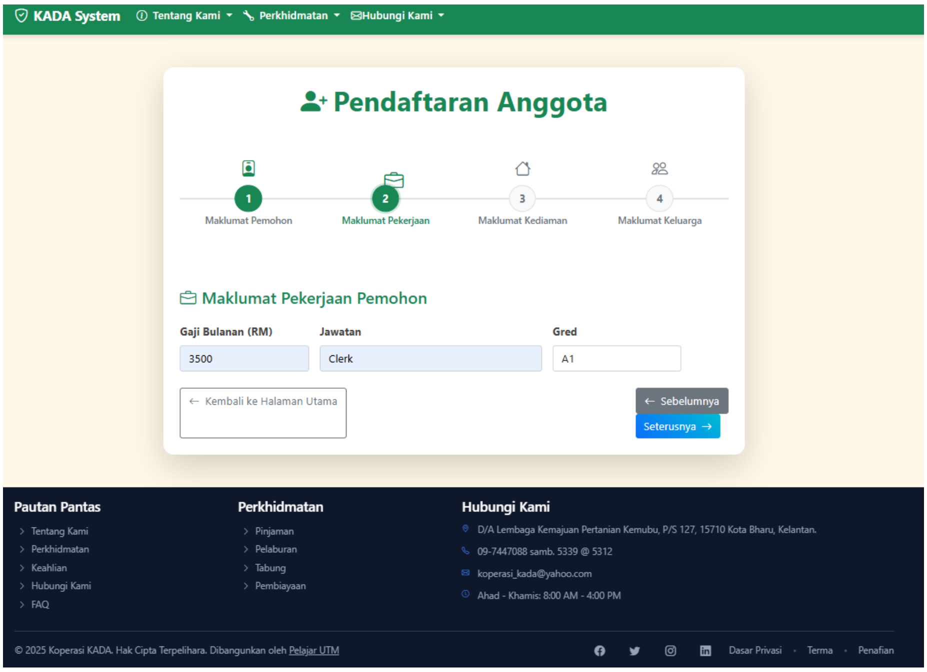
Output:

memberID	memberName	accountNumber	balance
2	JANET DOE	SA002	800

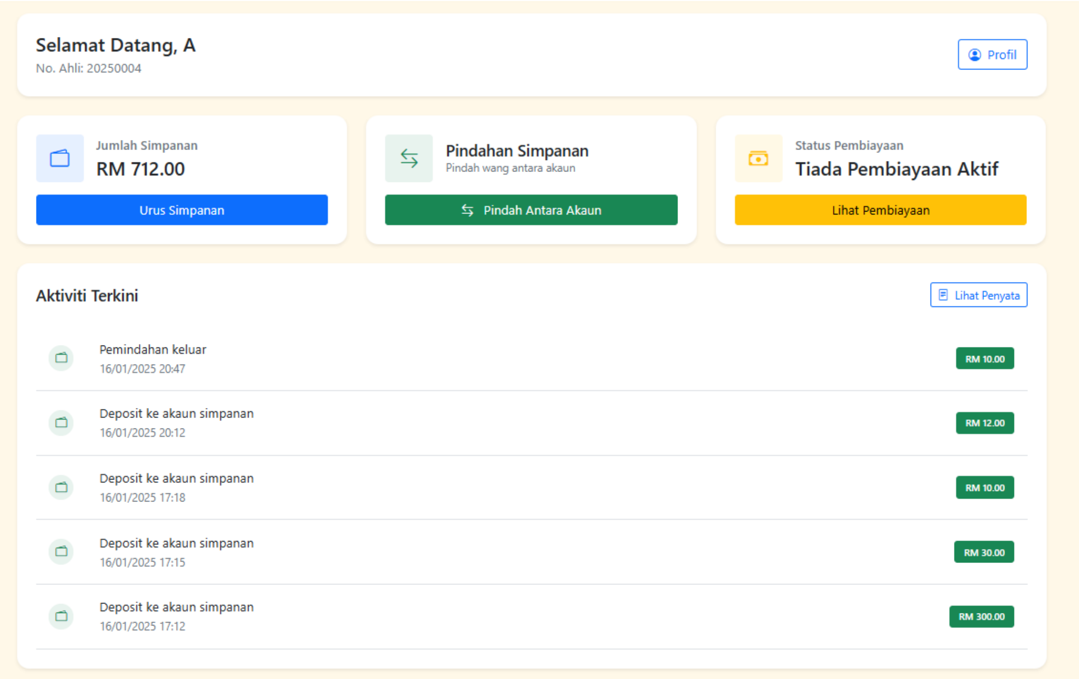
3.2 Interface Implementation

1) Member Registration Page

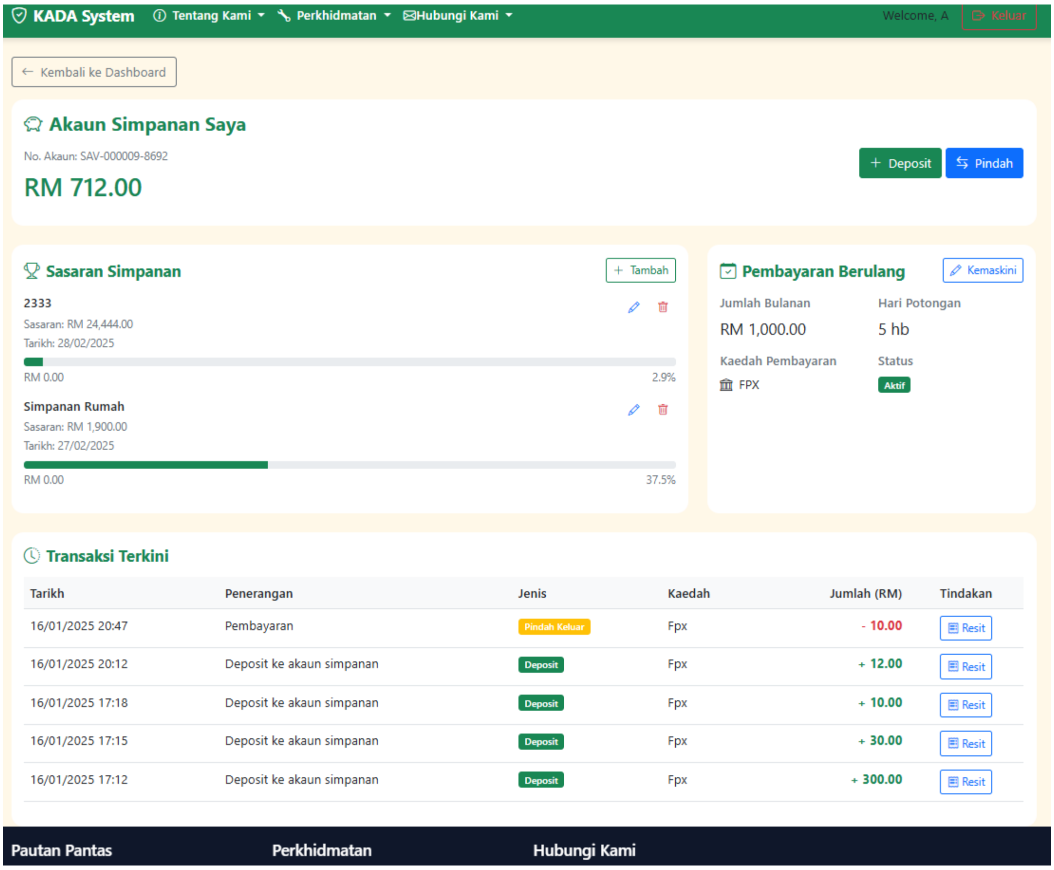




2) Account Page



3) Saving Dashboard Page



4) Transaction History

Transaksi Terkini

Tarikh	Penerangan	Jenis	Kaedah	Jumlah (RM)	Tindakan
16/01/2025 20:47	Pembayaran	Pindah Keluar	Fpx	- 10.00	Resit
16/01/2025 20:12	Deposit ke akaun simpanan	Deposit	Fpx	+ 12.00	Resit
16/01/2025 17:18	Deposit ke akaun simpanan	Deposit	Fpx	+ 10.00	Resit
16/01/2025 17:15	Deposit ke akaun simpanan	Deposit	Fpx	+ 30.00	Resit
16/01/2025 17:12	Deposit ke akaun simpanan	Deposit	Fpx	+ 300.00	Resit

SECTION D (CONCLUSION AND REFLECTION)

4.1 Conclusion

The Savings Module makes saving money easier and more convenient for both members and staff of the Koperasi KADA Online Platform. It has helpful features like automatic recurring savings, so members can save regularly without needing to do it manually. It also allows members to set and track their savings goals, helping them stay motivated. The dashboard is simple to use and shows all important information, such as savings, transactions, and goals, in one place.

These features make saving accessible and clear for members, while also reducing the workload for staff. Members can manage their savings anytime, while the system takes care of many tasks automatically, making the process smoother and more efficient for everyone.

In my opinion, the module can be improved further in the future. Adding a mobile app would let members manage their savings anywhere more easily. Besides that, adding stronger security features, like fingerprint or face login, would make accounts safer. Real-time notifications for savings and transactions could also keep members updated immediately, making the system even better.

4.2 Reflection

In the WBL project, our group worked on developing the Koperasi KADA Online Platform, a system designed to make cooperative processes more efficient. My task was to develop the Savings Module, which involved creating database structures, writing SQL queries, and designing user-friendly interfaces. Since each member in our group focused on a specific module, such as Login, Member Management, Loans Management, and Reports Management, teamwork and collaboration were essential to ensure that all the modules worked together smoothly.

At first, I felt overwhelmed by the complexity of designing the database. The idea of ensuring everything was normalized and properly structured seemed very difficult. However, with support from my team and guidance from our lecturer, I have become more confident on database as I progressed. Seeing the system gradually come together was a rewarding experience that made all the challenges worthwhile.

Throughout the project, I faced both positives and challenges. On the positive side, I gained a much better understanding of database normalization and learned how to implement SQL queries effectively. Working with my team also improved my problem-solving skills as we can solve issues together. However, there were also some challenges, such as ensuring that our designs aligned across all modules and that they integrated seamlessly into the overall system. These difficulties highlighted the need for careful planning and consistent communication.

This experience taught me the importance of modular design and teamwork in system development. The Savings Module, in particular, is an essential part of the platform as it provides tools for members to manage their finances efficiently. I realized how crucial it is to plan thoroughly and to work closely with others to achieve a shared goal.

Looking back, I discovered the value of structured planning, especially during the database design phase and module integration. Clear communication with my teammates was key to keeping the project on track. The question remains is how to manage resources and time to ensure efficiency of the system achieved.

In the future, I plan for an improvement by spending more time on testing and debugging to find potential issues. I also want to improve my skills in SQL optimization to make queries faster and more efficient. Additionally, I will seek feedback earlier in the design process to make improvements sooner.

In conclusion, this reflection has shown me areas where I can improve and reinforced the importance of learning step by step. It has been a valuable experience that will help me become better at handling similar projects in the future.